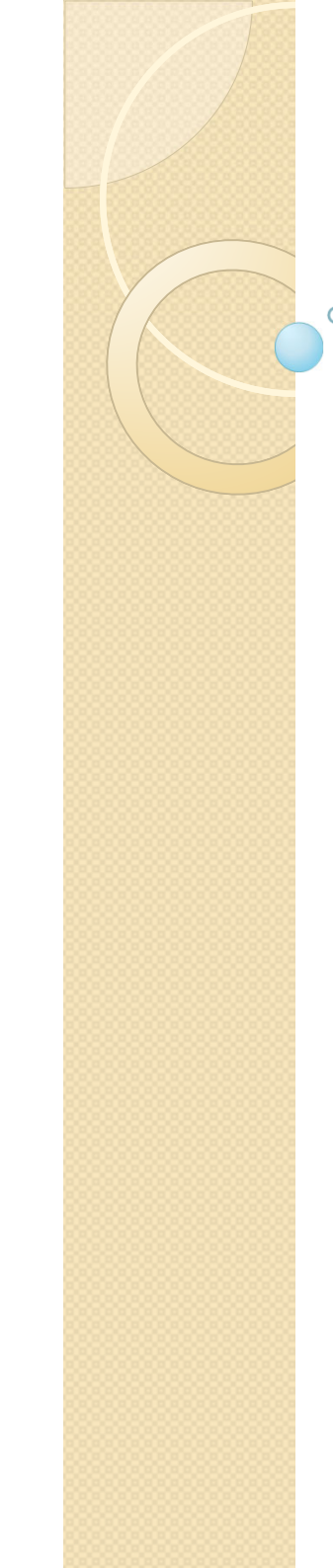




Project Based Learning Course

Class 1 and 2

2025/09/19

Kouzou Ohara

Outline of Class I

- Course Introduction
- Expected Social Issues in Thailand
- Related Technologies (AI, Bigdata, IoT)
- Dataset for Practice

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Japanese Lecturers

- Kouzou Ohara
Data Science, Information Science, and AI
(Machine Learning)
- Mako Shirouzu
Medicine, Healthcare, and Related Areas
- Yasuhiko Sugiyama
Organizer

Course Overview

- ✓ Gain hands-on experience in ***Monozukuri*** through application development using provided datasets
- ✓ Manage processes by defining roles and schedules within each group
- ✓ Conduct three presentations: conceptual design, system design, and final presentation
- ✓ Complete e-learning materials in advance (mandatory)

Requirements of This PBL Course

- It is mandatory to take the e-learning of this course in advance.
- It is mandatory to utilize the data provided for this PBL course to solve your problem.
Note that projects that do not use the provided data will be excluded from the evaluation.
- It is mandatory to use techniques related to AI or Machine Learning to solve your problem.
- Your solution should not involve developing any physical device because you can't develop and test it in this course.

Relationship between this course and the other courses

Technology Development Course

Learn how to consider new challenges or technologies required in the future through examples.

Ideas beyond existing technologies are important!

Technology Management Course

Learn the actual methods to develop a new product and manage its quality.

Career Development Course

Learn necessary matters to consider your own career path in the domain of *Monozukuri*.

Project Based Learning Course

Practice what you learn in the two courses using provided datasets and current technologies.

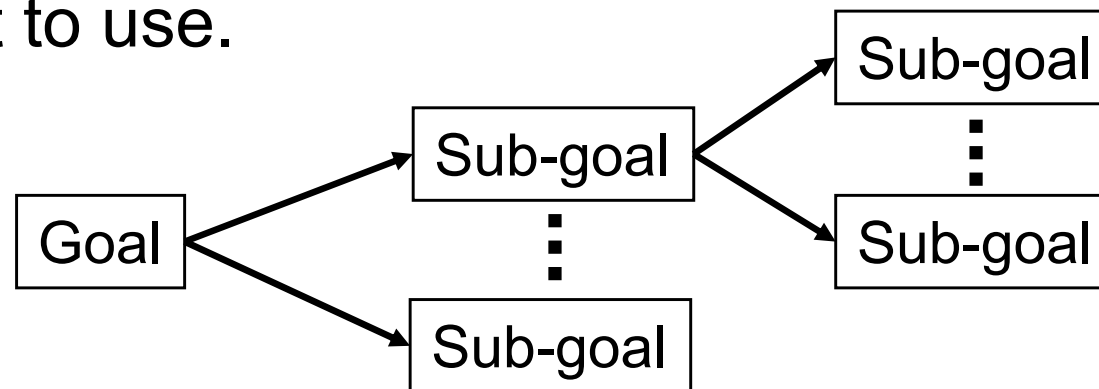
Goal-oriented thinking

Change your mind

from bottom-up/methodology-oriented
to top-down/goal-oriented thinking

Once you determine the goal of the project, divide it into sub-goals so that the goal can be achieved by achieving all of them. If necessary, further divide each sub-goal until you can directly solve it.

Don't start from thinking of methodologies you want to use.





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Recent Advances in AI

Generative AI and its applications are now commonly used in our daily life (ChatGPT, Gemini, etc.).



- ✓ Foundation models (large-scale deep learning models)
- ✓ Massive datasets for training (Big data)
- ✓ High-performance computing (GPU technologies)
- ✓ Cloud computing and distributed systems
- ✓ Scalable storage systems

AI Techniques Used in Applications

- Machine Learning
Prediction, Classification, Clustering, etc.
- Natural Language Processing
Interactive interface, Machine translation, etc.
- Computer Vision
Image/Scene recognition, Object detection, Image generation, etc.

AI Applications in the Real World

➤ On-demand type

e.g.) Translation apps, AI Chatbots such as ChatGP, Generative AI, Sales forecasting

- Activated only when needed
- Provides results on request

➤ Always-on Monitoring type

e.g.) Security surveillance, Industrial anomaly detection, **Patient monitoring**

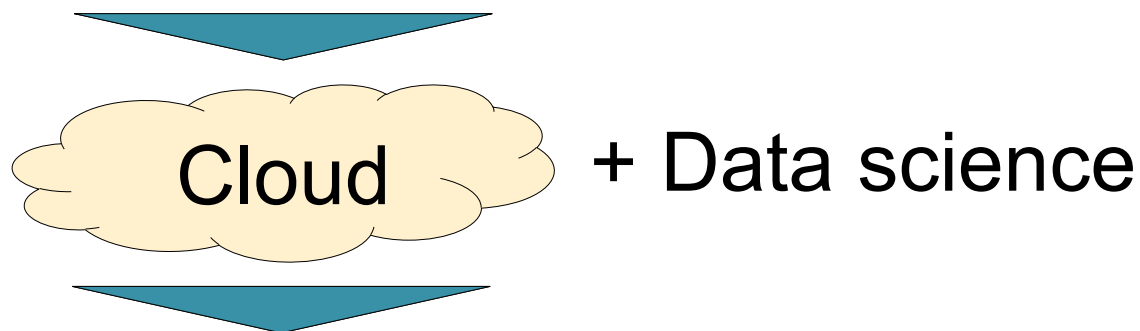
- Continuously running and monitoring
- Detects events or anomalies in real-time

How IoT Strengthens AI Applications

IoT (Internet of Things)

- Various devices connect to each other via the internet
e.g.) home appliances, cars, smart watch, etc.

Endless streams of data from IoT devices
(= observations of the real world)



- ✓ Expands the application field of AI
- ✓ Improves output accuracy and quality

Utilization in Healthcare

Sensing body/location information
by wearable devices



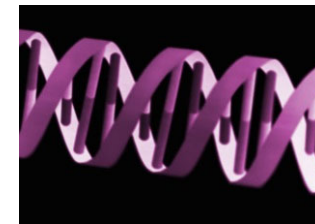
Wrist band type
device



Glass type
device

Vital sign

Inexpensive genetic test
(only specific genes)



Prediction and prevention
of specific diseases



Examination information

Utilizing Data in Medical domain

EBM: Evidence-Based Medicine

- Individual doctor's heuristics

- ➡ Clinical guidelines based on the analysis of clinical data

- Planned evaluation of a certain treatment

- ➡ Accumulation and analysis of large amounts of surgery and treatment data

Similar cases

- ➡ Assistance in decision making during treatment

Data Integration in Medical domain



Electronic medical record

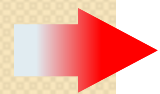
Detail information
of patients

Payment data

Partial data of the
treatment history
by each hospital

Medical prescription data

Information recorded on
prescription, such as
medicines, dosage,
and frequency
(no medical condition and
no laboratory data)



Enable to gather and analyze more detailed progress
information by tracking survey across hospitals
(risk assessment of symptoms, prediction of drug side effects)

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- **Dataset for Practice**

Overview of Dataset

Data of 53 subjects (41 in data_0 and 12 in data_1)

- Features of each subject
 - Personal attributes such as Gender, Age, etc.
 - Interview results on the degree of life independence
 - Results of the cognitive function test
 - Results of medical inspection
 - Evaluations by a medical doctor
- Sensing data of each subject
 - Time series data provided by the sensing device
 - Available for not all subjects
 - Variable sequence length

Data files

Files in the folder named data_ **X** (**X** = 0 or 1)

- **X**1_Subject_attributes
 - **X**2_Life_independence
 - **X**3_Cognitive_function
 - **X**4_Inspection_results
 - **X**5_evaluation
 - Meteorological_Data.xlsx
- } Excel format with headings
and
CSV format (UTF-8 with BOM)
without headings

➤ { Data_of_Ground_observatory.csv
Data_of_Residential.csv

- 71_sensing_data.zip in data_0 / sensing_data.zip in data_1

➤ cefoxSR* (data folder for each subject)

- cefoxSR*_activ_YYYYMMDDhhmmss.csv
- cefoxSR*_exist_YYYYMMDDhhmmss.csv
- cefoxSR*_sphy_YYYYMMDDhhmmss.csv
- cefoxSR*_temp_YYYYMMDDhhmmss.csv
- cefoxSR*_weight_YYYYMMDDhhmmss.csv



Class 2

Outline of Class 2

Conceptual design

- What you should think of in this class
- Overview of knowledge discovery from data
- Important things in data analysis
- Use of Generative AI



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Conceptual Design in System Development

The objective of this course is

To learn key aspects of real manufacturing through hands-on system development.

Conceptual design is

A crucial phase that determines whether a project is suitable for implementation.

What issue will you address?

Clearly answer to the following questions:

- ✓ Who are the stakeholders?
- ✓ Who are the users?
- ✓ What is your goal?
 - What are the inputs to your system?
 - What are the outputs of your system?
- ✓ What technologies do you need to achieve your goal?

What data do you need to solve the problem?

Do you have enough data for that purpose?

- * The provided data in this course are limited.

Therefore, it is necessary to choose a problem that can be solved solely with the provided data. However, you may also use additional publicly available data by collecting it yourself. In that case, you should clearly state the external data sources you intend to use.

Are there any missing values?

Are there any outliers?

How will you use the data?

Examples of possible ways to use the data you have:

- ✓ Train a machine learning model to generate predictive values
- ✓ Discover novel relationships or knowledge within the data that can lead to value creation

Note:

Simply visualizing the data is insufficient in this course, since it does not involve any AI-based techniques.



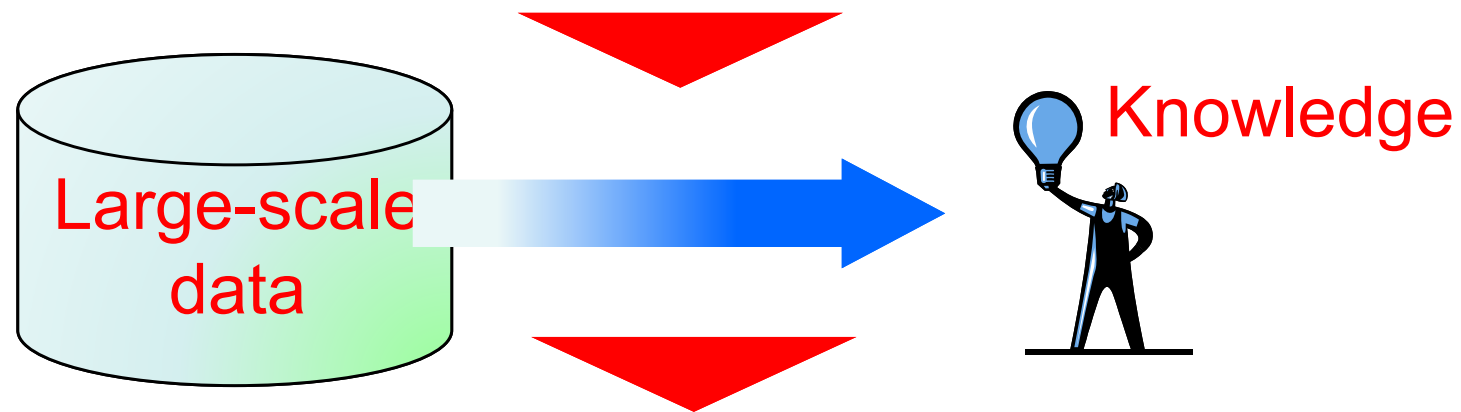
Outline of Class 2

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From Data to Knowledge

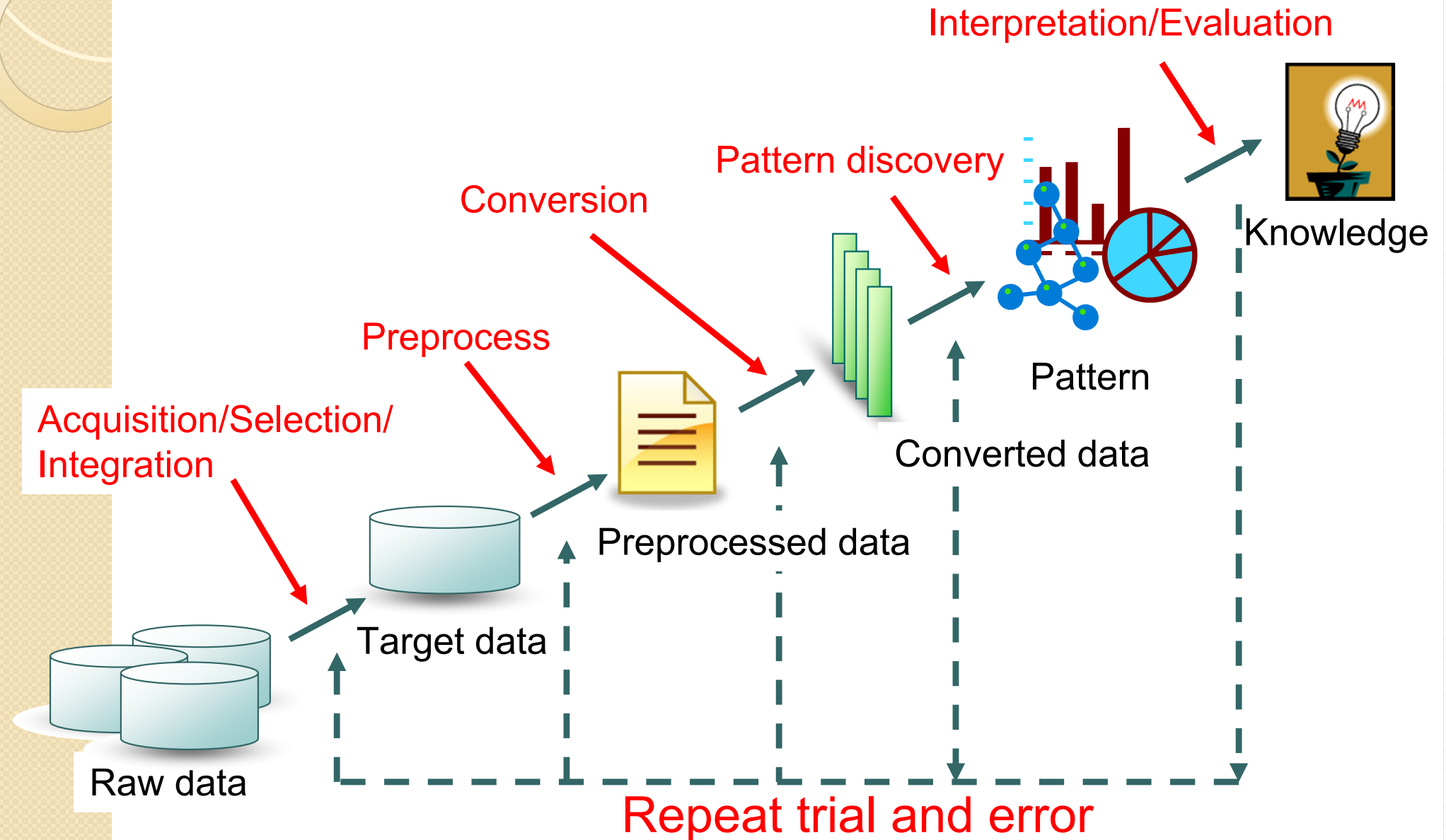
Raw data $\neq$ Knowledge



Knowledge Discovery from Database (KDD)
Data Science (Data Mining)

Techniques to efficiently find useful information
and knowledge from massive electronic data

Process of KDD



Techniques at each step

Acquisition/Selection/Integration

- Data warehousing

Preprocess

- Noise/exception handling
- Discretization of numerical attributes

Conversion

- Data format conversion
- Feature selection/construction

Pattern discovery ← Data mining in a narrow definition

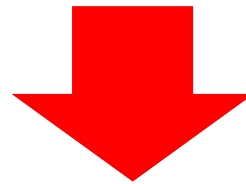
- Statistical method
- Machine Learning (AI)
- Methods specific to data mining

Interpretation/Evaluation

- Statistical method

Testing Machine Learning Model

If your solution employs a prediction/classification model that is based on a machine learning algorithm



Evaluate its prediction/classification accuracy with an **independent test dataset** in a proper way, such as **cross validation**. The evaluation of the model is important **to avoid over-fitting**.

Note that the evaluation results are not reliable if you use the test data when training the model, which is known as **data leakage (train-test contamination)**.



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What is important in data analysis?

What is the most important in data analysis is

Observing given data carefully before applying sophisticated tools.

You should get a deep insight into the data by observing it from various angles before determining your goal and developing your system.

Basic statistics

For numerical values

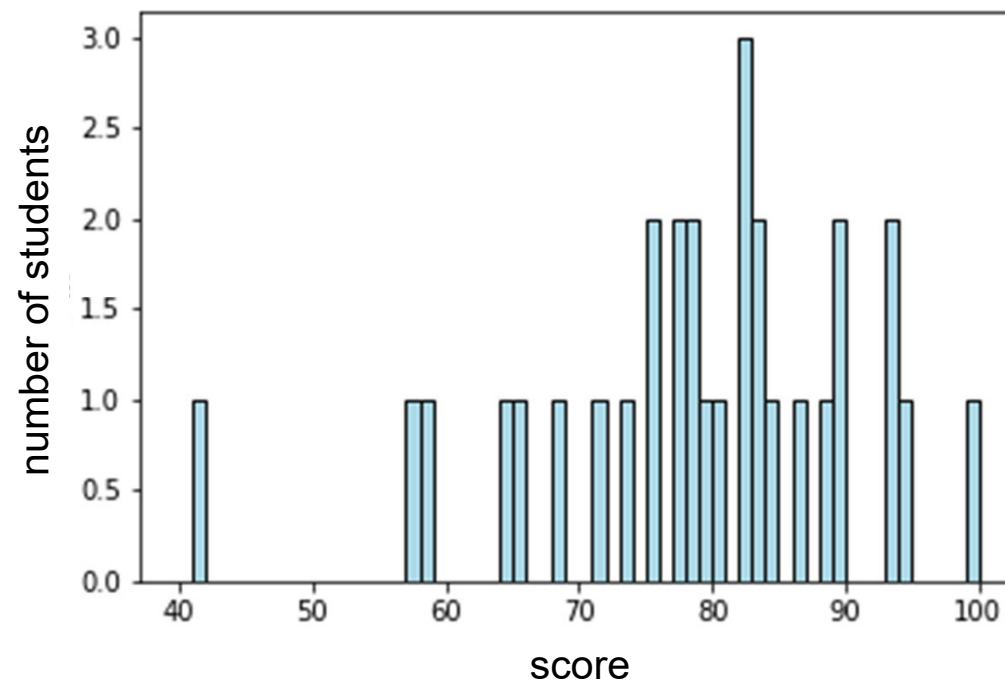
- Number of data points
- Maximum/Minimum values
- Mean/Median/Mode
- Range/Variance/Standard Deviation/
Inter Quartile Range
- Quantile/Percentile
- Correlation between two variables

For categorical values

- Variation of data (vocabulary)
- Frequency of each term (histogram)
- Co-occurrence relation

Visualizing Data - histogram -

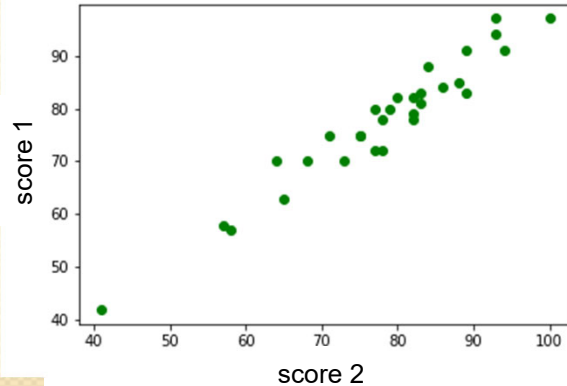
- Histogram
to know the distribution and bias of data



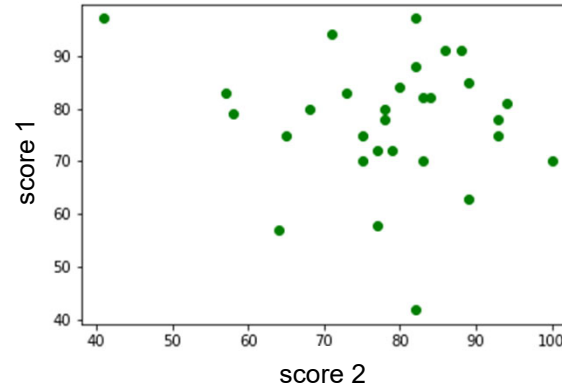
Visualizing Data - scatter plot -

- Scatter plot
to know the correlation between two variables

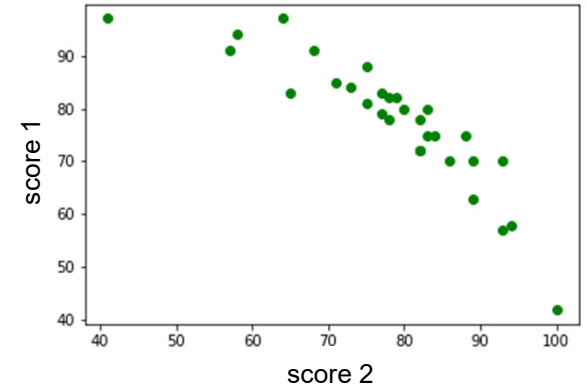
Positive



No correlation

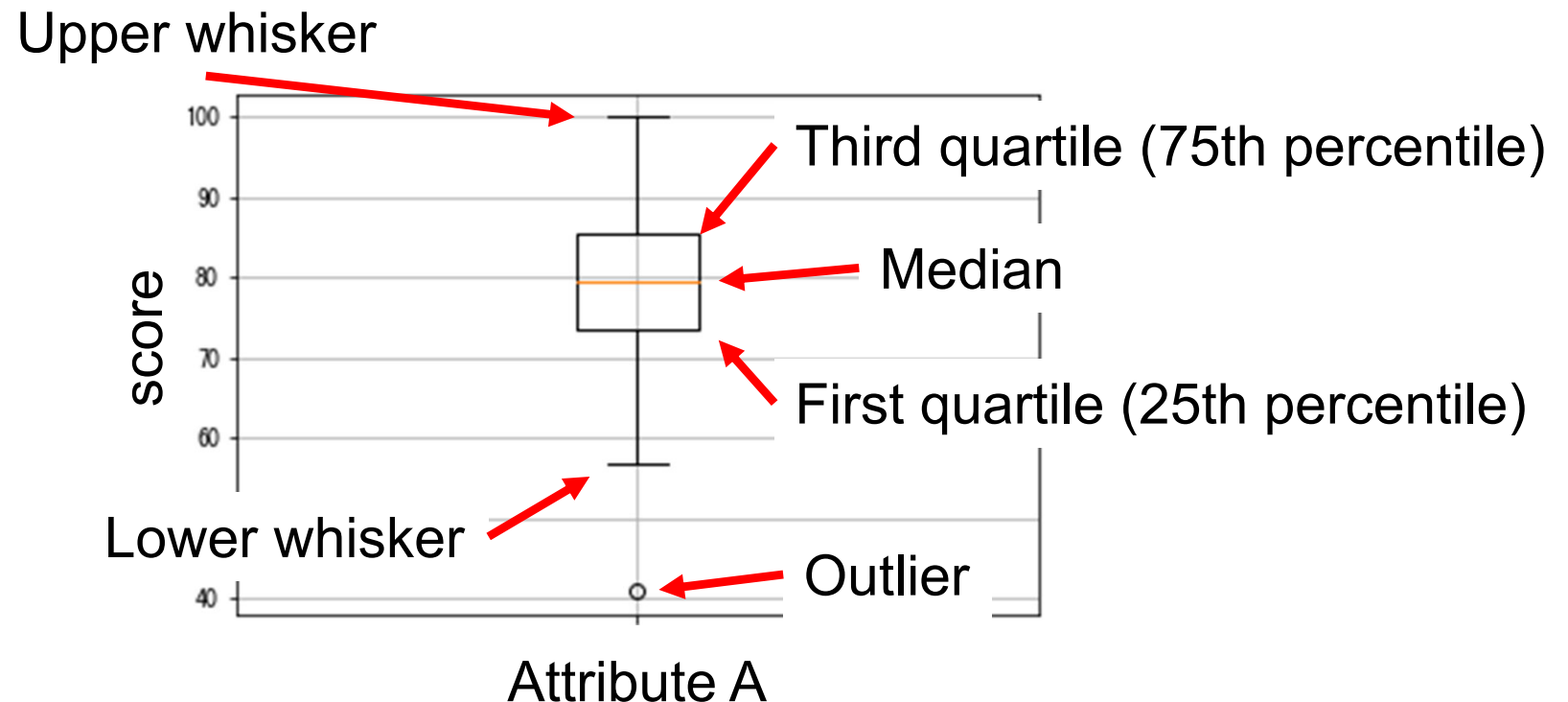


Negative



Visualizing Data - box plot -

- Box plot
to know the existence of outliers of a single feature
based on the five-number summary





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Embedding Generative AI into Your System

If sufficient computational resources are available to use a generative AI, you may use it in your proposed system.

However, note the following:

- ✓ The output of the generative AI is not always correct.
- ➡ You need to have a mechanism in place to provide the correct output to the user.
- ✓ Input data may be used to retrain the model.
- ➡ You may use generative AI only if the generative AI model you are using either runs in your local environment or is available through an API that allows you to limit the use of the data for re-training.

Medical / health data are privacy sensitive

Examples of Using Generative AI

- Chatbot interfaces for natural interaction

It is necessary to prepare a prediction/classification model separately. The chatbot would communicate its predictions to the user in natural language.

- Question and answer session by RAG*

Your system would generate supplemental descriptions for a predictive result from linguistic resources containing medical information using a generative AI model.

*RAG: Retrieval-Augmented Generation

Other Notes on the Use of Generative AI

- ✓ Follow the guidelines of the University.
- ✓ You must clearly state the originality of your proposal.
- ✓ The system should necessarily use the data provided.

For example, it is prohibited to create a chatbot using only a generative AI model without using the provided data.

- ✓ Pre-trained models are not suitable for forecasting.

When using generative AI as a predictive/classification model, additional training in the local environment is required.



Let's start thinking
together with
your group members!